Henery Garção

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RESEARCH PROFILE

Oceanographer and PhD candidate in Environmental Engineering (UFES). My work applies numerical models to marine, coastal and fluvial systems. A thread since the MSc is the Lagrangian framework — particle tracking of pollutants and floating objects — and its stochastic use to quantify risk. The PhD develops a probabilistic framework for flood risk estimation in data-scarce river basins, with application to territorial planning. Though distinct in application from my earlier coastal and marine work, the underlying logic is the same: quantifying environmental risk under conditions of incomplete information.

EDUCATION

PhD, Environmental Engineering — UFES, 2025 – present (supervisor: Prof. José Antônio Tosta dos Reis; FAPES scholarship). Probabilistic framework for flood risk assessment in data-scarce watersheds, integrating hydrological and hydraulic modelling with stochastic input sampling and uncertainty propagation, yielding probabilistic flood inundation maps for risk management applications.

MSc, Environmental Engineering — UFES, 2008 – 2010 (ANP scholarship). Development of a Lagrangian particle model incorporating wind effects and oil slick spreading.

BSc, Oceanography — UFES, 2004 – 2007 (CNPq scholarship). Hydrodynamics and salinity distribution in Vitória Bay through computational modelling (2DH numerical model).

PUBLICATIONS

Journal article. De Carvalho, G.V.; Cabral, M.M.; De Oliveira, A.L.S.C.; **Garção, H.F.**; Pellegrini, J.A.C. Methodology to Evaluate the Coastal Susceptibility to Oil Spills Originated in Large Marine Areas — Costa Norte Project. *Journal of Coastal Research*, 95(SI):1344–1348, 2020. doi:10.2112/SI95-258.1

Book chapter. Santos, F.A.; De Oliveira, A.L.S.C.; … **Garção, H.F.**; … Fragoso, M.R. Projeto Azul: Operational Oceanography in an Active Oil and Gas Area, Southeastern Brazil. In: *Coastal Ocean Observing Systems*, Elsevier, 2015, pp. 99–116. doi:10.1016/B978-0-12-802022-7.00020-1

Report chapter. Oliveira, A.L.S.C.; Carvalho, G.V.; **Garção, H.F.**; Pellegrini, J.A.C.; Cabral, M.M. Modelagem de Dispersão de Óleo — Capítulo 8. In: *Projeto Costa Norte — Desenvolvimento de Metodologias para o entendimento de processos costeiros e estuarinos e da vulnerabilidade de florestas de mangue na Margem Equatorial Brasileira*, v.2. Enauta/ANP, 2020.

CONFERENCE PRESENTATIONS

De Cindra de Souza, L.C.; **Garção, H.F.**; Silva, A.E.; Nascimento, T.F.; Manente Filho, A. Desenvolvimento de boias meteoceanográficas inteligentes para monitoramento em tempo real. Simpósio Brasileiro de Geologia e Geofísica Marinha, Vitória, 2025.

Cabral, M.M.; De Oliveira, A.L.S.C.; Alves dos Santos, F.; **Garção, H.F.**; De Carvalho, G.V. A stochastic simulation methodology for preliminary evaluation of environmental susceptibility to oil spills originated in large marine areas. Ocean Predict, Halifax, 2019.

De Oliveira, A.L.S.C.; Cabral, M.M.; **Garção, H.F.**; Pellegrini, J.A.C. Evaluation of the Interannual Variability in Stochastic Maps of Potential Oil Leakages within Exploratory Offshore Areas in the Equatorial Margin of Brazil. World Ocean Circulation User Consultation Meeting, Frascati, 2019.

Garção, H.F.; Cerrone, B.N.; dos Santos, F. Comparação de resultados de modelos numéricos e derivadores oceânicos na Bacia Sudeste Brasileira. XIV Congresso Latino-Americano de Ciências do Mar, Balneário Camboriú, 2011 (poster).

Garção, H.F.; Chacaltana, J.T.A. Estudo da hidrodinâmica e salinidade na Baía de Vitória (ES) através de modelagem computacional. XVIII Simpósio Brasileiro de Recursos Hídricos, Campo Grande, 2009.

Garção, H.F.; Chacaltana, J.T.A. Estudo da distribuição de salinidade na Baía de Vitória através de modelagem computacional. XIII Congreso Latinoamericano de Ciencias del Mar, Havana, 2009.

RESEARCH PROJECTS

PhD research — probabilistic flood risk for territorial planning (UFES, 2025–). Hydrological and hydraulic modelling with stochastic sampling of inputs and uncertainty propagation; probabilistic flood risk maps as the final output.

Projeto Costa Norte (2019–2022; Enauta · ANP · UFRJ · UFPA · UERJ). Technical coordinator, oil dispersion modelling; methodology for coastal vulnerability of mangroves in the Pará-Maranhão and Foz do Amazonas basins.

SPAD-SAR (Shell · Prooceano · Brazilian Navy; ANP PD&I; estimated R$ 7 million, 3 years). Project coordinator and programmer; Lagrangian drift modelling for search and rescue.

Projeto Azul (2013–2022; Shell Brasil · COPPE/UFRJ · Prooceano). Technical team member; ocean observation and forecasting system for the Santos Basin.

Smart meteoceanographic buoys (Umi San, 2025–). Contributor and co-author; telemetry, quality control and automation for real-time coastal monitoring.

PROFESSIONAL POSITIONS

Modelling Coordinator, Umi San — Hidrografia e Engenharia, 2025 – present

Senior Oceanographer, Fugro Middle East & India, Dubai, 2023 – 2024

Project Coordinator (2015 – 2022) and **Oceanographer** (2010 – 2015), CLS Brasil (formerly Prooceano)

Intern, Labesul, Department of Environmental Engineering, UFES, 2006 – 2007

SKILLS

Models: Delft3D, SWAN, MEMW (OSCAR, DREAM/ParTrack), HEC-HMS / HEC-RAS, CROCO (recently started), MOHID (course)

Programming and tools: Python, Fortran, Matlab, Git/GitHub, QGIS, ArcGIS, Surfer

TRAINING

Nature-based Solutions for Disaster and Climate Resilience — SDG Academy, online, 2024

Metocean Criteria and Fugro's Software — Fugro, Wallingford (training trip), 40 h, 2023

Coastal Systems Modelling with MOHID — University of São Paulo, 120 h, 2008

MEMW Training Course — SINTEF, Trondheim, 32 h, 2011

Fundamentals of Oil Spill Response; Advanced Oil Spill Modeling and Data Sources — International Oil Spill Conference, 2 × 2 h, 2014

Delivered: Oil Spill Dispersion Modelling Methodology, IBAMA licensing team, 2018 · OSCAR model, CLS headquarters, Toulouse, 24 h, 2016

LANGUAGES

Portuguese (native) · English (proficient) · Spanish (basic)